

Class 11: Modular Arithmetic

Goals:

Be able to construct Cayley tables for modular addition and modular multiplication.

Understand the ideas of closure, identity, and inverse elements.

Explore addition and multiplication tables mod 5, 6, 7, 8

$$\oplus_6$$

	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$\otimes_6$$

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2				
3	0	3				
4	0	4				
5	0	5				

$$\otimes_7$$

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2					
3	3					
4	4					
5	5					
6	6					

$$\otimes_7$$

	1	3	2	6	4	5
1	1	3	2	6	4	5
3	3					
2	2					
6	6					
4	4					
5	5					

The set $\mathbf{Z}_n$ with the operation of addition mod n has several nice features for every value of n .

The set is closed under the operation.

There is an additive identity.

Every element x has an additive inverse.

Every element shows up exactly once in every row and every column.

The set $\mathbf{Z}_n$ with the operation multiplication mod n is not as nice for all values of n . Which of the features is still true for multiplication?